

EVAN VIENNEAU

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University of Alberta, Edmonton, Alberta, Canada

RESEARCH INTERESTS

Dark matter, indirect detection, direct detection, quantum gravity, black holes

EDUCATION

Doctor of Philosophy (PhD) Physics

University of Alberta

September 2021 - Present

Edmonton, AB

- Using cosmological simulations to study the distribution and dynamics of dark matter halos in the context of direct and indirect detection, for a variety of astrophysical targets and dark matter models
- Using numerical methods to derive characteristics of quantum-corrected black holes modeling singularity resolution and dynamical horizons
- Anticipated completion in Spring 2026

Honours Bachelor of Physics Co-op

McMaster University

September 2016 - April 2021

Hamilton, ON

- Minor in computer science
- Graduated with an A-
- Senior thesis entitled "Properties of Low Mass Disk-Dominated Galaxies in Voids, Filaments, and Groups"

PUBLICATIONS

1. E. Pinetti, **E. Vienneau**, N. Bozorgnia, *Dark matter decay signals in cosmic filaments*, To be submitted to JCAP, [arXiv:2504.08025](https://arxiv.org/abs/2504.08025) [astro-ph.CO]
2. F. Fragomeno, D. M. Gingrich, S. Hergott, S. Rastgoo, **E. Vienneau**, *A generalized uncertainty-inspired quantum black hole*, Phys.Rev.D 111, 024048 (2025), [arXiv:2506.03909](https://arxiv.org/abs/2506.03909) [gr-qc]
3. **E. Vienneau**, A. J. Evans, O. V. Hartl, N. Bozorgnia, L. E. Strigari, A. H. Riley, N. Shipp, *Significant impact of Galactic dark matter particles on annihilation signals from Sagittarius analogues*, JCAP 10 (2024) 019, [arXiv:2403.15544](https://arxiv.org/abs/2403.15544) [astro-ph.HE]
4. E. Rahimi, **E. Vienneau**, N. Bozorgnia *The local dark matter distribution in self-interacting dark matter halos*, JCAP 02 (2023) 040, [arXiv:2210.06498](https://arxiv.org/abs/2210.06498) [astro-ph.CO]
5. T. Brown,..., **E.Vienneau**, et al., *VERTICO: The Virgo Environment Traced In CO Survey*, ApJS 257 21, [arXiv:2111.00937](https://arxiv.org/abs/2111.00937) [astro-ph.GA]

PRESENTATIONS

Impact of the LMC on DM indirect detection

2024 Canadian Association of Physicists Congress

May 2024

Western University

- First place student talk in particle physics division

A generalized uncertainty-inspired quantum black hole

Loops'24 International Conference on Quantum Gravity

May 2024

Florida Atlantic University

The local dark matter distribution in self-interacting dark matter halos

2023 Canadian Association of Physicists Congress

June 2023

University of New Brunswick

Enhanced star formation from ram-pressure stripping in Virgo galaxies
VERTICO Collaboration Meeting

December 2020
McMaster University

Properties of disk-dominated galaxies in voids, filaments, and groups
Department of Physics and Astronomy Poster Fair

May 2020
McMaster University

Calibration source umbilical behaviour in linear alkylbenzene
SNO+ Bi-Yearly Collaboration Meeting

March 2019
SNOLAB

SCIENTIFIC VISITS

2023 Canadian Astroparticle Physics Community Meeting
QueensU

August 2023
Kingston, Ontario

- Attended a variety of presentations, focused on communicating the research progress achieved by the Canadian astroparticle physics community

2022 Tri-Institute Summer School on Elementary Particles
TRIUMF

July 2022
Vancouver, British Columbia

- Attended a diverse array of lectures to learn, engage and network with leading experts in the field

WORK EXPERIENCE

Astrophysics Researcher and Beta Tester
McMaster University

September 2020 - April 2021
Hamilton, ON

- Acted as beta-tester for the development of a state-of-the-art cloud-based computing environment named ARCADE
- Worked with an ALMA imaging pipeline to optimize the processing of interferometric radio continuum data from the VERTICO survey of galaxies
- Analysis of this data lead to potential evidence for enhanced radio continuum emission on the leading edge of infalling Virgo cluster galaxies

Physics Pedagogy Researcher
McMaster University

July 2020 - August 2020
Hamilton, ON

- Developed web-based Python programming bootcamp for incoming science undergraduates
- Investigated ways to better incorporate computation into the undergraduate physics curriculum
- Supported instructors in the organizational and logistical side of running online courses

Neutrino Detection Researcher
SNOLAB

January 2019 - June 2019
Lively, ON

- Ran quality assurance tests on scintillator samples using density, turbidity and UV absorption meters
- Monitored state of cover gas bags in detector cavity to maintain correct nitrogen levels
- Worked detector operator shifts monitoring and gathering detector data
- Performed independent research on the performance of calibration source umbilicals in scintillator fluid

OTHER RESPONSIBILITIES

Graduate Student Physics Association - Ambassador
University of Alberta

January 2024 - December 2024
Edmonton, AB

- Act as the main liaison between the graduate student physics association and the graduate student association

Graduate Student Physics Association - VP Finance
University of Alberta

January 2023 - December 2023
Edmonton, AB

- Oversee GPSA finances and manage GPSA bank account
- Act as signing authority
- Support fundraising activities

Physics Graduate Student Association - YUGSA representative
York University

September 2021 - December 2021
Toronto, ON

- Networking and community development

TEACHING ASSISTANTSHIP

PHYS144 - Newtonian Mechanics
University of Alberta

September 2024 - December 2024
Edmonton, AB

- Running and marking of lab sessions

PHYS146 - Fluids and Waves
University of Alberta

January 2024 - April 2024
Edmonton, AB

- Overseeing and facilitating semester long student research projects in a variety of physics subfields

ENPHYS131 - Mechanics
University of Alberta

January 2023 - April 2023
Edmonton, AB

- Running and marking of lab sessions

PHYS130 - Wave Motion, Optics and Sound
University of Alberta

September 2022 - December 2022
Edmonton, AB

- Running of large class tutorial sessions and marking

PHYS124 - Particles and Waves
University of Alberta

September 2022 - December 2022
Edmonton, AB

- Running of large class tutorial sessions and marking

PHYS1421 - Physics with Life Science Applications 1
York University

May 2022 - June 2022
Toronto, ON

- Invigilation of midterms and exams
- Marking of class assignments, midterms and exam

PHYS2030 - Computational Methods for Physicists and Engineers
York University

January 2022 - April 2022
Toronto, ON

- Marking of weekly assignments

PHYS1422 - Physics with Life Science Applications 2
York University

January 2022 - April 2022
Toronto, ON

- Marking of class tests

PHYS1411 - Physics Fundamentals 1
York University

September 2021 - December 2021
Toronto, ON

- Running of lab tutorial sessions and marking labs